AI-Powered Intelligent Document Retrieval System

Abstract

The rapid growth of digital information has created challenges in locating relevant knowledge efficiently. Traditional keyword-based search systems often fail to understand user intent and semantic meaning. This proposal presents an AI-powered document retrieval system that combines Retrieval-Augmented Generation (RAG), vector embeddings, and Large Language Models to provide context-aware responses.

Objectives

- Build a semantic document search system.
- Generate embeddings for uploaded documents.
- Store vectors in a vector database.
- Retrieve relevant context using similarity search.
- Generate accurate responses using an LLM.

Methodology

1. Upload PDF documents.
2. Extract text.
3. Split into chunks.
4. Generate embeddings.
5. Store vectors.
6. Retrieve relevant chunks.
7. Send context to LLM.
8. Return final answer.

Technology Stack

Backend

- FastAPI

Frontend

- React

Vector Database

- ChromaDB

Embedding Model

- Sentence Transformers

LLM

- Gemini / OpenAI / Llama

Programming Language

- Python

Expected Outcomes

- Fast semantic search
- Context-aware answers
- Reduced hallucinations
- Improved information retrieval
- Easy scalability

Applications

- Research assistants
- Educational platforms
- Enterprise knowledge bases
- Healthcare document search
- Legal document analysis
- Customer support systems

Future Work

- Multilingual support
- Voice-based querying

- Hybrid search
- Fine-tuned domain-specific LLMs
- Multi-modal document retrieval

Conclusion

The proposed system leverages modern AI technologies to improve document retrieval by understanding semantic meaning rather than relying solely on keyword matching. It provides accurate, scalable, and context-aware responses suitable for enterprise and research applications.